Python Network Scanner Project

1) Build Your First Python Port Scanner

Step 1 - Create Your Project Folder

Create a folder on your computer.

Example:

```
Cybersecurity-Projects
    └── network-scanner
```

Inside it create a file:

```
scanner.py
```

Step 2 - Write Your First Scanner Code

Open the file and paste this code:

scanner.py - C:/Users/asus/Cybersecurity-Projects/network-scanner/scanner.py (3.14.3)

File Edit Format Run Options Window Help

```python
import socket

target = "google.com"
port = 80

scanner = socket.socket(socket.AF_INET, socket.SOCK_STREAM)

result = scanner.connect_ex((target, port))

if result == 0:
    print("Port is OPEN")
else:
    print("Port is CLOSED")

scanner.close()
```

Step 3 - Run the Script

Open Terminal.

Navigate to the folder: cd network-scanner

Run: python scanner.py

Output:

```
PS C:\Users\asus\Cybersecurity-projects> cd network-scanner
PS C:\Users\asus\Cybersecurity-projects\network-scanner> python scanner.py
Port is OPEN
PS C:\Users\asus\Cybersecurity-projects\network-scanner>
```

2) Build a Multi-Port Scanner

Step 1 - Update Your Script

Replace your code with this:

scanner.py - C:/Users/asus/Cybersecurity-Projects/network-scanner/s...

File Edit Format Run Options Window Help

```python
import socket

target = "google.com"

print("Scanning target:", target)

for port in range(1, 101):

    scanner = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
    scanner.settimeout(1)

    result = scanner.connect_ex((target, port))

    if result == 0:
        print(f"Port {port} is OPEN")

    scanner.close()
```

Step 2 - Run the Scanner

Open terminal and run: python scanner.py

Your script will now check **ports 1-100**.

Output:

```
>>>
    === RESTART: C:/Users/asus/Cybersecurity-I
    Scanning target: google.com
    Port 80 is OPEN

>>>
```

3) Make the Scanner Interactive (User Input)

Step 1 - Update Your Code

Replace your current code with this:

scanner.py - C:/Users/asus/Cybersecurity-Projects/network-scanner/s...

File Edit Format Run Options Window Help

```python
import socket

target = input("Enter target (IP or website): ")
start_port = int(input("Enter start port: "))
end_port = int(input("Enter end port: "))

print(f"\nScanning target: {target}\n")

for port in range(start_port, end_port + 1):

    scanner = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
    scanner.settimeout(1)

    result = scanner.connect_ex((target, port))

    if result == 0:
        print(f"Port {port} is OPEN")

    scanner.close()
```

Step 2 - Run the Scanner

In terminal run: python scanner.py

Output:

```
=== RESTART: C:/Users/asus/Cybersecurity-Projects/network-scanner/scanner.py ===
Enter target (IP or website): facebook.com
Enter start port: 1
Enter end port: 100

Scanning target: facebook.com

Port 80 is OPEN
>>>
```

4) Make the Scanner Faster with Multithreading

Step 1 - Import Thread Library

Add this at the top of your script:

```python
from threading import Thread
```

Step 2 - Updated Code

Replace your script with this:

scanner.py - C:/Users/asus/Cybersecurity-Projects/network-scanner/s...

File Edit Format Run Options Window Help

```python
import socket
from threading import Thread

target = input("Enter target (IP or website): ")
start_port = int(input("Enter start port: "))
end_port = int(input("Enter end port: "))

print(f"\nScanning target: {target}\n")

def scan_port(port):

    scanner = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
    scanner.settimeout(1)

    result = scanner.connect_ex((target, port))

    if result == 0:
        print(f"Port {port} is OPEN")

    scanner.close()


for port in range(start_port, end_port + 1):

    thread = Thread(target=scan_port, args=(port,))
    thread.start()
```

Step 3 - Run the Scanner

Run: python scanner.py

Output:

```
>>>
    === RESTART: C:/Users/asus/Cybersecurity-Projects/network-scanner/scanner.py ===
    Enter target (IP or website): www.google.com
    Enter start port: 1
    Enter end port: 200

    Scanning target: www.google.com

    Port 80 is OPEN
>>>
```

Now ports will scan **much faster**.

5) Detect Running Services on Open Ports

Step 1 - Create a Service Dictionary

Add this near the top of your code:

scanner.py - C:\Users\asus\Cybersecurity-Projects\network-scanner\scanner.py (3.14.3)

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```
    21: "FTP",
    22: "SSH",
    23: "Telnet",
    25: "SMTP",
    53: "DNS",
    80: "HTTP",
    110: "POP3",
    139: "NetBIOS",
    143: "IMAP",
    443: "HTTPS"
}
```

This maps **port numbers → services**.

Step 2 - Update the Scan Function

Modify your scan_port function.

```
def scan_port(port):

    scanner = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
    scanner.settimeout(1)

    result = scanner.connect_ex((target, port))

    if result == 0:

        service = services.get(port, "Unknown")

        print(f"Port {port} OPEN ({service})")

    scanner.close()
```

Step 3 - Run the Scanner

Example run: python scanner.py

Output:

```
=== RESTART: C:\Users\asus\Cybersecurity-Projects\network-scanner\scanner.py ===
Enter target (IP or website): google.com
Enter start port: 1
Enter end port: 500

Scanning target: google.com

Port 80 OPEN (HTTP)
Port 443 OPEN (HTTPS)
>>>
```

```
=== RESTART: C:\Users\asus\Cybersecurity-Projects\network-scanner\scanner.py ===
Enter target (IP or website): scanme.nmap.org
Enter start port: 1
Enter end port: 500

Scanning target: scanme.nmap.org

>>> Port 22 OPEN (SSH)Port 80 OPEN (HTTP)
```

6) Final Upgrade: Make the Scanner Portfolio-Ready

Step 1 - Add a Banner

At the top of your script add:

scanner.py - C:\Users\asus\Cybersecurity-Projects\network-scanner\scanner.py (3.14.3)

File Edit Format Run Options Window Help

```
print("""
====================================
      PYTHON NETWORK SCANNER
      Cybersecurity Project
====================================
""")
```

Output:

```
=== RESTART: C:\Users\asus\Cybersecurity-Projects\network-scanner\scanner.py ===

====================================
      PYTHON NETWORK SCANNER
      Cybersecurity Project
====================================

Enter target (IP or website):
```

Step 2 - Create a Result File

Add this before scanning starts:

Python

```python
output_file = open("scan_results.txt", "w")
```

This creates a file where results will be stored.

Step 3 - Save Scan Results

Update the scan function:

```python
def scan_port(port):

    scanner = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
    scanner.settimeout(1)

    result = scanner.connect_ex((target, port))

    if result == 0:

        service = services.get(port, "Unknown")

        message =(f"Port {port} OPEN ({service})")

        print(message)
        output_file.write(message + "\n")

    scanner.close()
```

Step 4 - Close the File

At the end of the script add:

```python
output_file.close()
```

Output:

```
    ====================================
          PYTHON NETWORK SCANNER
          Cybersecurity Project
    ====================================

    Enter target (IP or website): www.google.com
    Enter start port: 1
    Enter end port: 100

    Scanning target: www.google.com

    Port 80 OPEN (HTTP)
>>>
```